

Part 2: Product Review

Part A: Product Review & Gap Analysis

The PRD solves a real problem for customers who waste time doing manual reports. The flow makes sense, especially the option to save a report right from the data tables. However, I found a few gaps we should fix before development starts, mainly around permissions, what happens when data changes, and export limits.

Here are the top 5 gaps I found, prioritized by their impact:

1. Report Permissions vs. Delivery (Critical)

What is unclear: The PRD mentions RBAC and organization-level sharing, but it doesn't explain what happens if a user schedules a report to be sent to a colleague who has lower permissions.

Why it matters: Reports often contain sensitive data. If a manager with high-level access schedules an automated report to be sent to a junior employee, we need to know if the system checks the junior employee's permissions. If it doesn't, the system might accidentally leak sensitive company data to unauthorized users via email.

To close the gap: We need a clear product rule. The engineering and security teams must define whether the emailed report is generated according to the creator's permissions or filtered strictly by the recipient's permissions.

2. Handling Deleted Dashboards (High)

What is unclear: A PDF Summary Report can be synced to a Custom Dashboard so that it always reflects the latest changes. However, the PRD does not address what happens to the scheduled report if the original dashboard is heavily edited or completely deleted.

Why it matters: If an automated workflow relies on a source that no longer exists, the system might crash, or worse, send an empty or broken report to an executive. This breaks the automation cycle and creates a frustrating experience.

To close the gap: We need to define a fallback mechanism. For example, if a source dashboard is deleted, the system should automatically pause the scheduled workflow and send a notification email to the report creator so they can fix it.

3. The 500K Row Limit in CSV Exports (High)

What is unclear: The CSV creation mockup states that CSV supports up to 500K records, but the behavior at or above that limit is not specified.

Why it matters: If the system silently cuts off the data at 500K rows without telling the user, a security analyst might assume they are looking at the full picture when they are actually missing critical data. Large exports also take a long time and could cause the system to time out.

To close the gap: We need to define exactly what happens at this limit. The system should ideally complete the export up to the limit, but clearly display a warning message inside the file stating that the data was truncated and suggesting the user apply more filters.

4. The UX Contradiction for Large Files (High)

What is unclear: The PRD states that if a file is too large for an email attachment, the system will send an authenticated download link instead. But it does not define the size limit or how the authentication works.

Why it matters: The PRD specifically mentions that Summary Reports (PDFs) are meant for executives so they can read the data without logging into the platform. Forcing an executive to click a link, remember a password, and log in just because a file is slightly too big contradicts the main goal of the feature. They will likely ignore the report.

To close the gap: Define the exact attachment size limit. We should also explore better user experiences for executives, such as sending a One-Time Password (OTP) to their email instead of requiring a full platform login.

5. Operational Data Time Semantics (Medium-High)

What is unclear: Summary Reports explicitly support a relative time range, but the Operational Data flow does not clearly define time behavior. It is unclear whether a CSV represents the current matching findings, historical findings, findings first/last seen in a period, or another model.

Why it matters: Without a clear time filter in the CSV creation flow, users might accidentally request to download years of historical data instead of just the recent issues. This will drastically increase the chances of hitting the 500K row limit and overloading our databases.

To close the gap: Product and Design need to update the mockups to ensure that the time-selection step is included in the CSV workflow, just like it is in the PDF workflow.

Part B: Test Plan

The goal of this test plan is to ensure the core export pipeline works smoothly and that the system handles the edge cases identified in the product review safely.

ID	Test Case Description	Expected Result	Priority
T1	Standard PDF Generation: Create a Summary Report from a Custom Dashboard. Apply specific scope filters and a time range.	A readable PDF is successfully generated. The data perfectly matches the filters and timeframe, with no extra data included.	Critical
T2	Standard CSV Generation: Create an Operational Data CSV from a specific module. Reorder the columns in the settings.	The CSV downloads successfully, containing only the selected columns in the exact custom order requested.	Critical
T3	Contextual Creation Shortcut: Navigate to an existing data table, apply several filters, and click "Save as Custom Report".	The report creation menu opens smoothly with all previously selected filters and modules pre-populated, saving the user time.	High
T4	Sync Feature with Changes: Create a report linked to a Custom Dashboard with "Sync board changes" ON. Alter a graph on the original dashboard.	The next scheduled report automatically updates to reflect the changes made to the original dashboard.	High
T5	Deleted Source Error Handling: Schedule an automated report, then entirely delete the dashboard it is based on. Wait for the scheduled time.	The system handles the error without crashing. It pauses the schedule and sends a clear error notification to the report creator's email.	High
T6	Permissions Test: Try to schedule a report containing restricted data to a user account with lower-tier permissions.	The system enforces RBAC rules. It either blocks the delivery or filters out the restricted data so the recipient only sees what they are allowed to see.	Critical
T7	CSV Size Limit Test: Request an export of a massive dataset that exceeds the 500K row limit.	The system does not crash or time out. It stops at the defined limit and includes a clear warning for the user that the data was truncated.	High
T8	Large File Delivery: Generate a report with a file size that is intentionally larger than standard email attachment limits.	The user receives an email containing a secure download link instead of an attachment, and the link functions correctly upon authentication.	Critical

Explicitly Out of Scope for This Round

I will focus strictly on the reporting mechanism (data extraction, formatting, and delivery). Testing the actual accuracy of the security findings (e.g. verifying if a flagged vulnerability is truly a risk) is out of scope for this plan, as I assume the underlying security modules are already tested and functioning correctly.

Part C: BI & Monitoring Definition

After launch, the dashboard should answer two questions: (1) are customers adopting Custom Reporting and getting repeat value from it, and (2) is the system reliably generating and delivering the right reports? I would separate product/adoption metrics from operational-health metrics so a healthy service is not mistaken for a successful product.

1. Adoption & Usage Metrics

- **Active Organizations:** The number of unique organizations that created at least one report in the last 30 days. This is the baseline metric for feature adoption.
- **Contextual Creation Rate:** The percentage of reports initiated via the "Save as Custom Report" button inside data tables versus reports created from scratch on the main Reports page. This tells us if embedding the feature directly in the workflow is driving usage.
- **Automation Ratio:** The percentage of total reports that are connected to an active scheduled workflow. A high ratio proves the feature is successfully reducing repetitive manual work for our users.
- **Report Type Distribution:** The breakdown between PDF and CSV generations. This helps us understand our user base better - whether they are mostly managers needing visual summaries or technical users analyzing raw data.

2. Quality & Reliability Metrics

- **Generation Success Rate:** The percentage of all report requests that are completed successfully without errors. This is our primary service health indicator.
- **Scheduled Delivery Success Rate:** The percentage of scheduled emails that successfully reach the recipient. Generating a report is useless if the email fails to send.
- **Broken Source Rate:** The number of scheduled workflows that failed to run because the source dashboard was deleted or became inaccessible.
- **Export-Limit Hit Rate:** How often users try to export CSVs larger than the 500K row limit. If this happens frequently, we will need to reconsider the limit or improve data filtering options.

3. Alerting Strategy

Initial thresholds should be treated as starting points and tuned after we have a production baseline. I would avoid paging on isolated customer/configuration errors and page only when the signal suggests a platform-level issue.

- **Immediate Page to Engineering:** Trigger an alert if the Generation Success Rate drops below 95% for more than 15 consecutive minutes. This indicates a platform-level issue that needs urgent attention.
- **Weekly Product Review:** Review the Broken Source Rate and Export-Limit Hit Rate on a weekly basis to identify trends, understand user frustration points, and prioritize future UI improvements.

4. Data / Instrumentation Needed

To build the dashboards mentioned above I would work with the developers to implement event tracking. We need to measure user actions.

- **Funnel Events:** report_create_started, report_saved, and report_create_failed. This helps us build a conversion funnel to see where users drop off during the creation process.
- **Generation Events:** generation_started, generation_succeeded, and generation_failed (including the failure reason, the file size, and how long it took to generate).
- **Delivery Events:** delivery_succeeded, delivery_failed, and whether the report was sent as a direct attachment or a secure link.
- **Edge Case Events:** export_limit_reached (when they hit the 500K boundary) and source_missing (when a dashboard is deleted).
- **Engagement:** I want to look at actual clicks on the download link instead of tracking if the email was opened.
- **Privacy Note:** All product analytics should use internal anonymous IDs. We must never log the customers' actual email addresses or the security data from the reports into our general BI tools.

Suggested Dashboard Layout

I would organize the final BI dashboard into four clear sections so the product and engineering teams can quickly understand the data:

- **Adoption Panel:** Overall usage, active organizations, and the PDF vs. CSV mix.
- **Creation Funnel:** A step-by-step view from create flow started to report saved to identify if the UI is confusing.
- **Automation Panel:** Scheduled report stats and delivery success rates.
- **Reliability Panel:** The technical health check, showing Generation Success Rate, average generation time, 500K limit hits, and top errors.